WALLCALENDAR USER MANUAL

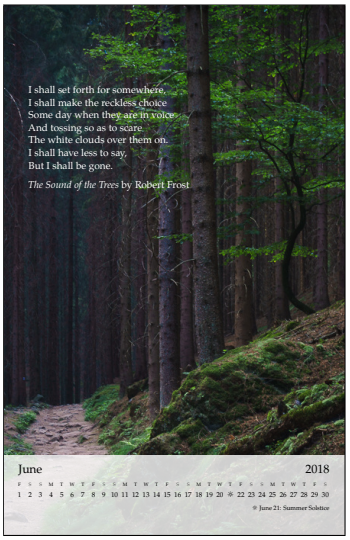
v1.6.0 · 14th March 2026 · link

This is the **User Manual** for the `wallcalendar` class. Clone or download from Github:

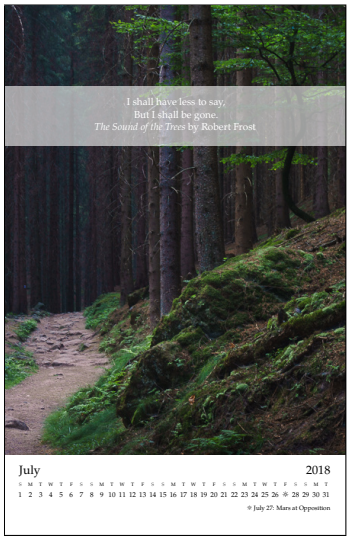
https://github.com/profound-labs/wallcalendar/

The documentclass comes with the following layouts:

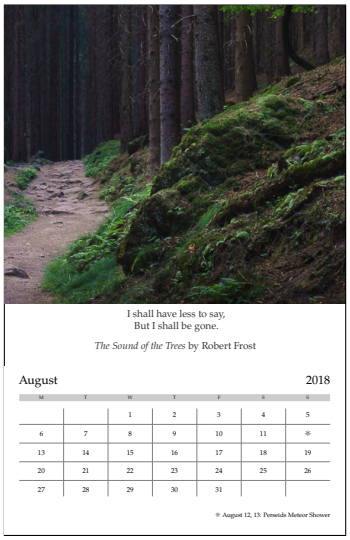
Full page photo, the calendar days overlaid with opacity



Full page photo, the photo above the calendar days



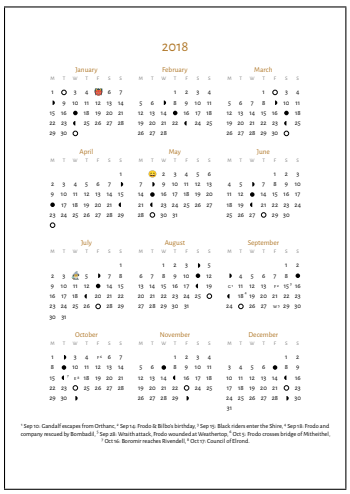
Small landscape photo, with a calendar grid



Title page

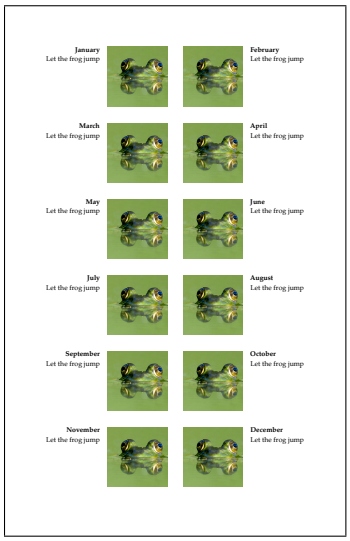
Photo and notes on two pages

Year planner

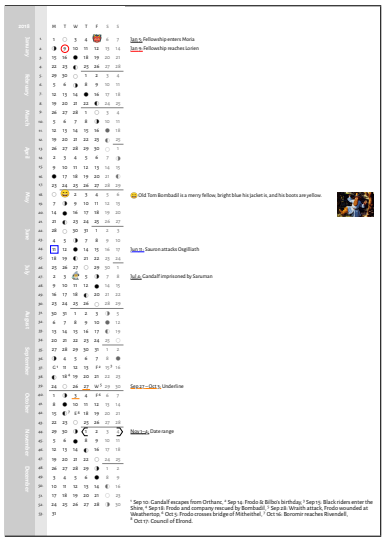


Load event marks from CSV file

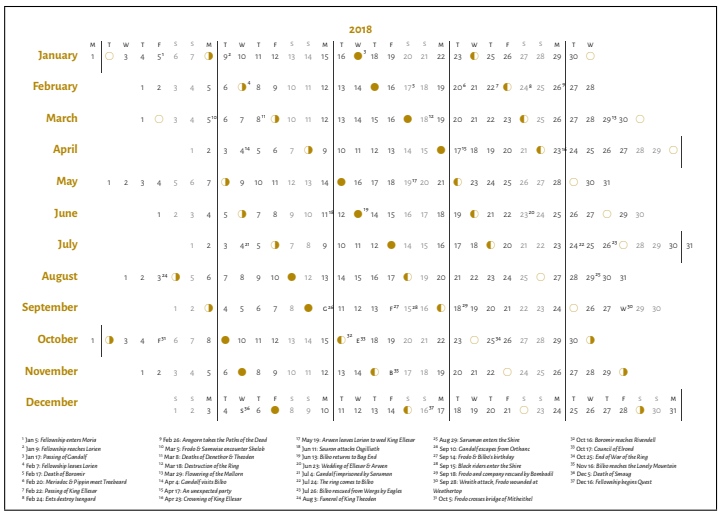
Thumbnails and captions

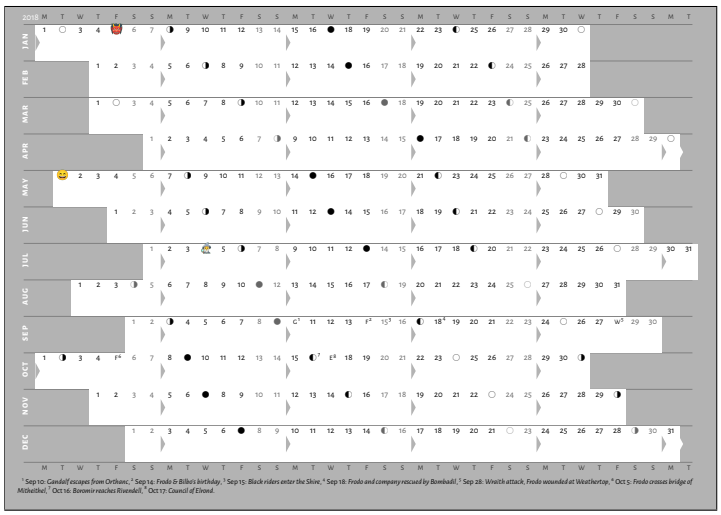


Compact year planner with weekly notes



Landscape year planners





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1 TUTORIAL: FOREST CALENDAR

In this tutorial we will produce the three example pages seen in the summary.

Set the parameters of the month pages in advance, either in the preamble or in the document body, but before calling `\MonthPage{ month }` to typeset it.

A month page can have four areas:

- Photo
- Quote
- Calendar
- Events

Their parameters are set separately for each month:

```
\SetPhoto[ options ]{ month }
\SetQuote[ options ]{ month }{ quote text }
\SetCalendar[ options ]{ month }
\SetEvents[ options ]{ month }{ calendar tikz marks }{ events text }
```

The month page will be typeset with:

```
\MonthPage[ options ]{ month }
```

1.1 Documentclass

To start, load the documentclass and set `year`, `language` and the `imageFolder`:

```
\documentclass[
  year=2018,
  language=english,
  imageFolder=./photos/,
]{wallcalendar}
```

Let's start the preamble with `\makeatletter` to be safe.

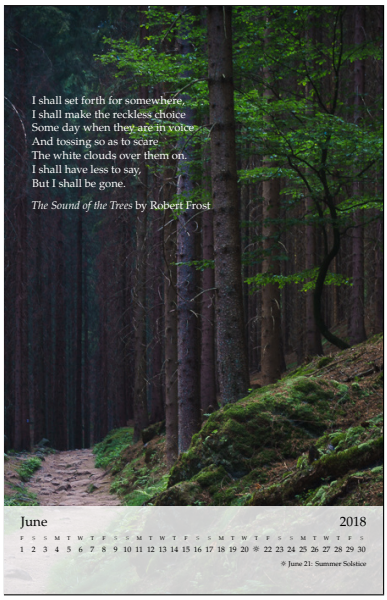
```
\makeatletter
```

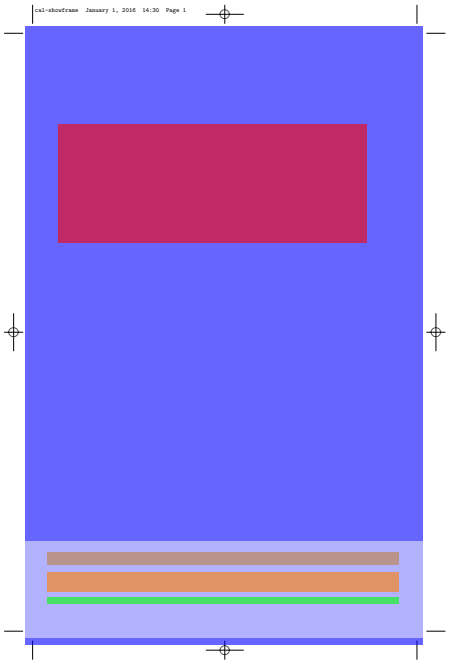
1.2 Font settings

For this example we'll use TEX Gyre Pagella as the main typeface. We also load DejaVu Sans to use a particular glyph as a mark in the calendar (`U+263C` white sun with rays).

```
\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}
```

1.3 June





`showtrims` and `showframe` class options show the page structure.

It will be a full page photo, with 3mm bleed on all four sides. You can see the bleed if you enable the `showtrims` class option. We also specify the file name of the photo (no extension), this will be the argument of `\includegraphics`.

```
\SetPhoto[bleed=3mm, file={obscure-crop}]{June}
```

A quote will be positioned over the photo. The quote is in a `\linewidth` wide minipage, attached to the top left corner of the page. Use `\raggedleft`, `\raggedright`, or `\centering` for alignment, and the `xOffset` and `yOffset` options to move the quote's minipage to the exact position.

```
\SetQuote[xOffset=-5mm, yOffset=-20mm]{June}{%
\raggedright
\setlength{\parskip}{10pt}%
\Large
\color{white}

I shall set forth for somewhere,\\
I shall make the reckless choice\\
Some day when they are in voice\\
And tossing so as to scare\\
The white clouds over them on.\\
I shall have less to say,\\
But I shall be gone.

\textit{The Sound of the Trees} by Robert Frost
}
```

The layout macro will place the calendar at the bottom, dates in a single line.

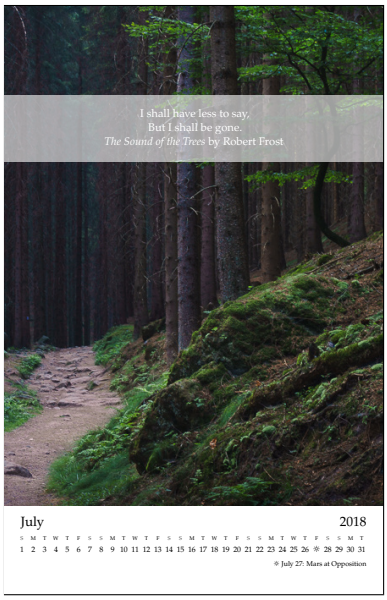
Here we use a conditional to use a different calendar style when `showframe` is turned on, this helps with debugging or tuning the position.

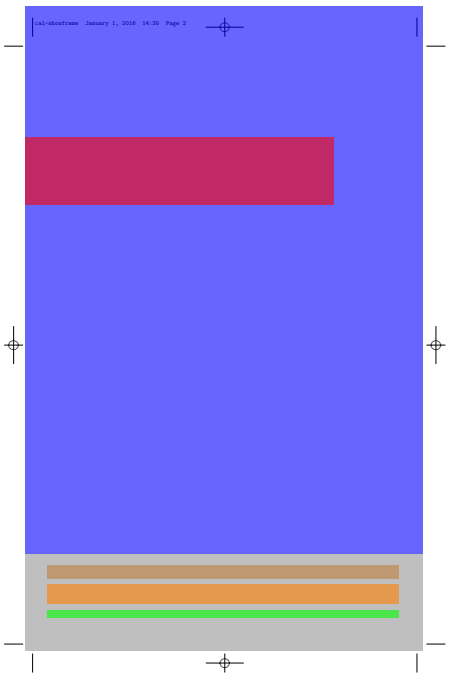
```
\ifshowframe
  \SetCalendar[bg/.style={opacity=0.5, fill=white}]{June}
\else
  \SetCalendar[bg/.style={opacity=0.8}]{June}
\fi
```

Events for particular days are printed under the calendar.

```
\SetEvents{June}{%
  if (equals=2018-06-21)
    [day text={\dejaVuSans\char"263C}];% U+263C white sun with rays
}{%
\raggedleft
{\dejaVuSans\char"263C} June 21: Summer Solstice
}
```

1.4 July



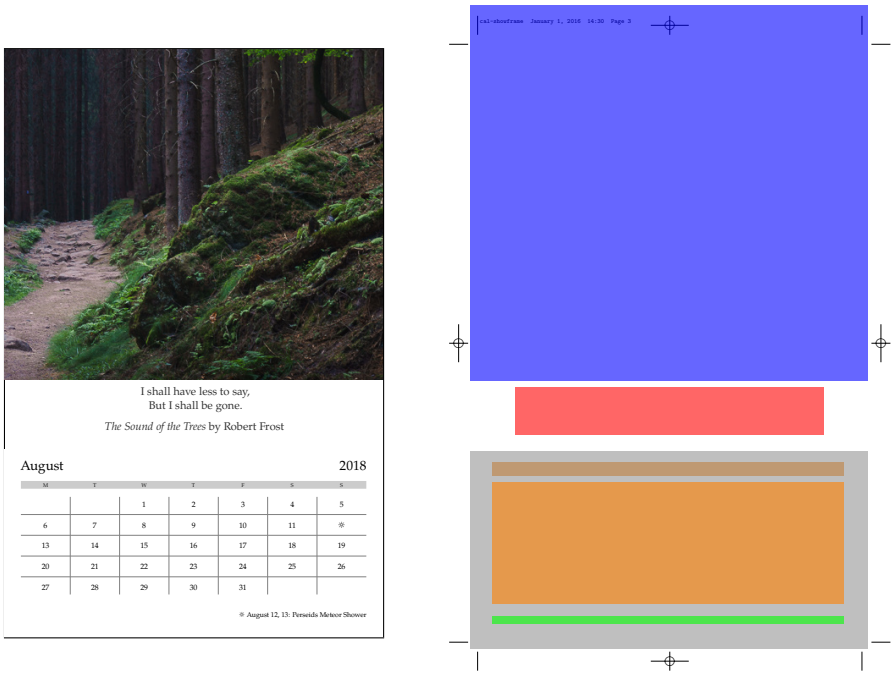


Same as June, but we will set the image to be placed above the calendar, and we add a transparent background for the quote.

This layout is a good option when the top or the bottom of the photo has to be cropped, and you can't use the full page aspect ratio for the photo.

```
\SetPhoto[bleed=3mm, file={obscure-crop}]{July}

\SetQuote[%
  xOffset=0.5\linewidth - 0.5\paperwidth -3mm,
  yOffset=-20mm,
]{July}{%
\begin{tikzpicture}%
\node [
  fill=white, opacity=0.6, minimum width={\paperwidth + 3mm},
  minimum height=30mm] {};%
\node [] {%
\begin{minipage}{\paperwidth + 3mm}%
\centering
\Large
\color{white}

I shall have less to say,\\
But I shall be gone.

\textit{The Sound of the Trees} by Robert Frost
\end{minipage}%
};
\end{tikzpicture}%
}

\ifshowframe
  \SetCalendar[bg/.style={opacity=0.5}]{July}
\else
  \SetCalendar[bg/.style={opacity=1}]{July}
\fi

\SetEvents{July}{
  if (equals=2018-07-27) [day text={\dejaVuSans\char"263C}];
}{%
\raggedleft
{\dejaVuSans\char"263C} July 27: Mars at Opposition
}
```

1.5 August



This layout works for photos that are horizontal (landscape orientation), scaled into the bleed margin on three sides.

```
\SetPhoto[bleed=3mm, file={obscure-crop}, yOffset=-150mm]{August}

\SetQuote[yOffset=-3mm]{August}{%
\centering
\setlength{\parskip}{10pt}%
\Large
\color{black!80}

I shall have less to say,\\
But I shall be gone.

\textit{The Sound of the Trees} by Robert Frost
}

\ifshowframe
  \SetCalendar[bg/.style={opacity=0.5}]{August}
\else
  \SetCalendar[bg/.style={opacity=1}]{August}
\fi

\SetEvents{August}{
  if (equals=2018-08-12) [day text={\dejaVuSans\char"263C}];
}{%
\raggedleft
{\dejaVuSans\char"263C} August 12, 13: Perseids Meteor Shower
}
```

End of the preamble.

```
\makeatother
```

1.6 The document

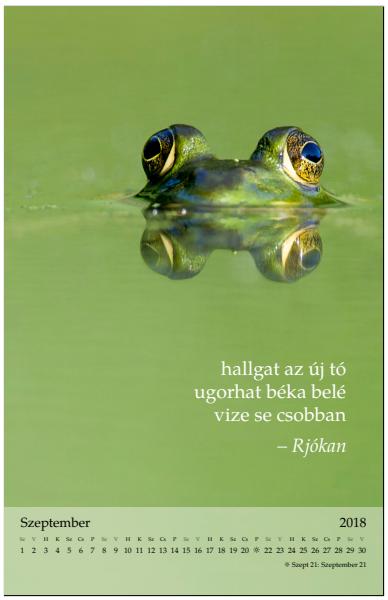
Typesetting the month pages in the document is now just this much:

```
\begin{document}

\MonthPage[layout=full page, put photo=full page]{June}

\MonthPage[layout=full page, put photo=full width above calendar]{July}

\MonthPage[layout=small landscape, put photo=full width]{August}

\end{document}
```

2 TUTORIAL: TRANSLATIONS

In this tutorial we will produce the same calendar in three languages: Japanese, English and Hungarian.

We are going to use `IPAPMincho` font for the Japanese.



2.1 Files

The main document files:

```
cal-translations-japanese.tex
cal-translations-english.tex
cal-translations-hungarian.tex
```

Fonts, formatting settings, etc.:

```
local-japanese.sty
local-english.sty
local-hungarian.sty
```

Translation text input:

```
frog-japanese.tex
frog-english.tex
frog-hungarian.tex
```

Setup month pages (same across translations):

```
frog.tex
```

2.2 Translations setup

Create the `frog-english.tex` file and use the `\SetTxt{ key }{ content }` command to set text content for translation keys.

`frog-japanese.tex`

```
\SetTxt{September Quote}{%
{\mincho       }

{\Large\textit{araike ya / kawazu tobikomu / oto mo nashi}}

{\mincho  }%
}

\newcommand\SeptMarks{%
  if (equals=2018-09-21)
    [day text={\dejaVuSans\char"263C}];% U+263C white sun with rays
}

\SetTxt{Sept Events}{%
{\dejaVuSans\char"263C} {\mincho   21 :      21 }
}
```

`frog-english.tex`

```
\SetTxt{September Quote}{%
silent new lake\\
let the frog jump\\
not even a sound

\textit{-- Ryokan}%
}

\newcommand\SeptMarks{%
  if (equals=2018-09-21)
    [day text={\dejaVuSans\char"263C}];% U+263C white sun with rays
}

\SetTxt{Sept Events}{%
{\dejaVuSans\char"263C} Sept 21: September 21
}
```

`frog-hungarian.tex`

```
\SetTxt{September Quote}{%
hallgat az új tó\\
ugorhat béka belé\\
vize se csobban

\textit{-- Rjókan}%
}

\newcommand\SeptMarks{%
  if (equals=2018-09-21)
    [day text={\dejaVuSans\char"263C}];% U+263C white sun with rays
}
```

```
\SetTxt{Sept Events}{%
{\dejaVuSans\char"263C} Szept 21: Szeptember 21
}
```

NOTE: Using \SetTxt{} to store values intended as tikz marks on the calendar will not work. The \txt{} command will be the value of \@eventmarks and tikz can't resolve it there.

Put the calendar marks in a command instead, as above with \SeptMarks.

```
\calendar (cal#1)
  [alnitak, dates=\CalendarYear-#1-01 to \CalendarYear-#1-last]
  \@eventmarks;%
```

```
% NOTE This code below will not work.
% Put the calendar marks in a command instead.

\SetTxt{Sept Marks}{%
  if (equals=2018-09-21)
    [day text={\dejaVuSans\char"263C}];% U+263C white sun with rays
}

% ...

\SetEvents{September}{%
\txt{Sept Marks}
}{%
\raggedleft
\txt{Sept Events}
}
```

2.3 Document setup

Load the documentclass. We are setting the translations option to define the file where translation keys are set. This file is loaded by the documentclass as an \input.

cal-translations-japanese.tex

```
\documentclass[
  year = 2018,
  language = japanese,
  translationsInputFile = frog-japanese.tex,
  imageFolder = ./photos/,
]{wallcalendar}

\usepackage{local-japanese}

% Content is the same across translations
\input{./frog.tex}

\begin{document}

% Just one month
\MonthPage[layout=full page, put photo=full page]{September}

\end{document}
```

local-japanese.sty

```
\ProvidesPackage{local-japanese}

\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}
% Japanese font
\newfontfamily\mincho{IPAPMincho}

% Renew formatting hooks to use the \mincho font
\renewcommand\fullPageFmt{%
  \renewcommand*\monthFmt{\LARGE\mincho}%
  \renewcommand*\yearFmt{\LARGE\mincho}%
  \renewcommand*\dayLetterColor{}%
  \renewcommand*\dayLetterFmt{\tiny\mincho}%
  \renewcommand*\dayTextFmt{\small}%
  \renewcommand*\quoteFmt{}%
  \renewcommand*\headingFmt{\centering}%
  \renewcommand*\calendarFmt{\centering}%
  \renewcommand*\eventsFmt{%
   \setlength{\parindent}{0pt}\raggedleft\footnotesize%
  }%
}
```

cal-translations-english.tex

```
\documentclass[
  year = 2018,
  language = english,
  translationsInputFile = frog-english.tex,
  imageFolder = ./photos/,
]{wallcalendar}

\usepackage{local-english}

\input{./frog.tex}

\begin{document}

\MonthPage[layout=full page, put photo=full page]{September}

\end{document}
```

local-english.sty

```
\ProvidesPackage{local-english}

\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}
```

cal-translations-hungarian.tex

```
\documentclass[
  year = 2018,
  language = hungarian,
  translationsInputFile = frog-hungarian.tex,
  imageFolder = ./photos/,
]{wallcalendar}

\usepackage{local-hungarian}

\input{./frog.tex}

\begin{document}

\MonthPage[layout=full page, put photo=full page]{September}

\end{document}
```

local-hungarian.sty

```
\ProvidesPackage{local-hungarian}

\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}
```

frog.tex

```
\makeatletter

\SetPhoto[bleed=3mm, file={frog-crop}]{September}
```

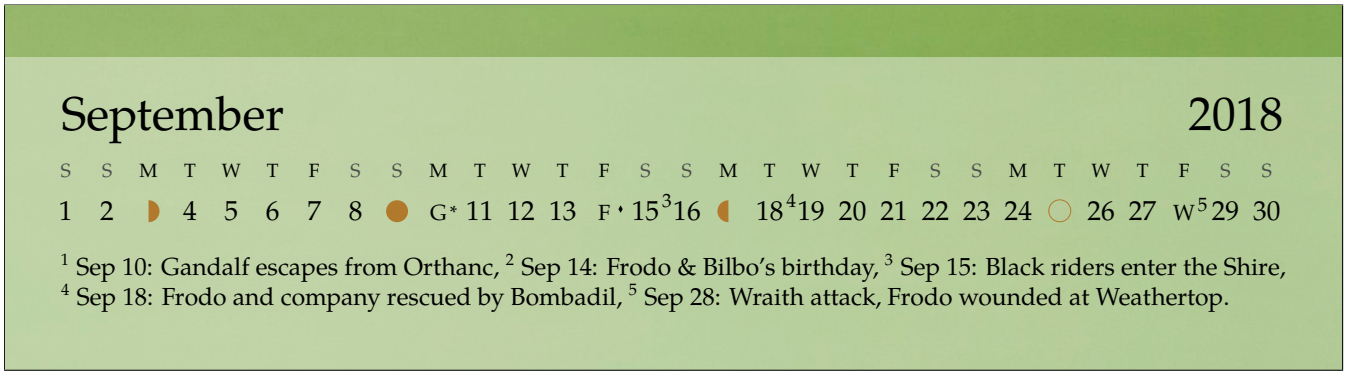
Use the `\txt{ key }` command to load text from translation keys:

```

\SetQuote[xOffset=0pt, yOffset=-140mm]{September}{%
\raggedleft\setlength{\parskip}{10pt}\HUGE\color{white}%
\txt{September Quote}%
}
```

Calendar settings for the month, using `\txt` to access translated parts.

```
\SetCalendar[bg/.style={opacity=0.4}]{September}

\SetEvents{September}{%
\SeptMarks%
}{%
\raggedleft
\txt{Sept Events}
}

\makeatother
```

3 TUTORIAL: LOAD EVENTS FROM CSV



3.1 CSV FILES

Events in the CSV should be already sorted by date.

If you are using more than one CSV, put all events with notes (i.e. indexed entries) in the same CSV. The index number of the mark is taken from the row number in the CSV, so a second CSV with notes would start the count from 1 again.

We're going to use the following csv files, see in the `./doc/examples/data/` folder.

`holidays.csv`

`moonphases.csv`

`mark_defaults.csv`

3.2 EVENT FORMATTING

You can format the event output by setting the `format cmd` key:

```
\parseMonthEvents[%
  format cmd = {%
    \textsuperscript{\eMark}~\eMonthShort~\eDay:\space%
    \eNote\ifnumless{\eIdx}{\eMaxIdx}{,\space}{.}%
  },
]%
```

Or define a Lua formatting function and set it with the `format func` key:

`helpers.lua`

```
function eventFmtCustom(idx, max_idx, event, event_date, mark)
  local d = event_date
  tex.sprint(string.format(
    "\\textsuperscript{%s} & %s %s: & %s \\\\",
    mark.symbol, d:fmt("%b"), d:getday(), event.note
  ))
end
```

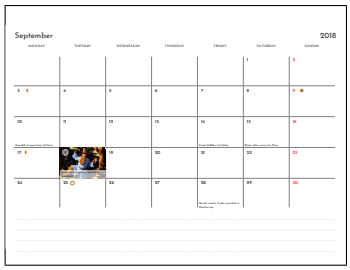
```
\parseMonthEvents[format func = eventFmtCustom]%
```

3.3 Document setup

cal-marks.tex

```
\documentclass[
  year = 2018,
  eventsCsv = ./data/holidays.csv,
  markDefaultsCsv = ./data/mark_defaults.csv,
  imageFolder = ./photos/,
]{wallcalendar}

\makeatletter

\colorlet{mooncolor}{darkgold}

\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}

\SetPhoto[bleed=3mm, file={frog-crop}]{September}

\SetQuote[xOffset=0pt, yOffset=-140mm]{September}{%
\raggedleft\setlength{\parskip}{10pt}\HUGE\color{white}%
silent new lake\\
let the frog jump\\
not even a sound

\textit{-- Ryokan}%
}

\SetCalendar[bg/.style={opacity=0.4}]{September}

\SetEvents{September}{%
\parseMonthMarksDayTextUsing{./data/moonphases.csv}%
\parseMonthMarksDayText%
\parseMonthMarksNote%
}{%
\raggedright
\parseMonthEvents[%
  format cmd = {%
    \textsuperscript{\eMark}~\eMonthShort~\eDay:\space%
    \eNote\ifnumless{\eIdx}{\eMaxIdx}{,\space}{.}%
   },
]%
}

\makeatother

\begin{document}

\MonthPage[layout=full page, put photo=full page]{September}

\end{document}
```

4 EXAMPLE: PHOTO AND NOTES WITH TITLE PAGE



See `doc/examples/cal-photo-and-notes.tex`.

4.1 Title page

A title page with a full page photo can be produced with:

`\TitlePage{photo path}{overlay height}{title text}`

Crop the photo to the aspect ratio of the trimmed paper size.

Open the image in GIMP, and set “Fixed aspect ratio” on the Rectangle Select tool. It takes integers, so you can use the page dimensions multiplied up. For US Letter (11 x 8.5 inch), this would be `110:85`. Then select the area and crop.

It also helps to put guides on the edges by selecting all (Ctrl + A) and using Image > Guides > New Guides from Selection.

4.2 Adding background photos to the calendar days

This can be done by positioning a `\node` over the day.

The background photo has to be cropped to the aspect ratio of a grid cell in the calendar.

You can determine the aspect by first creating a calendar PDF with the intended page size, then open the PDF in GIMP. Import the calendar page at 300dpi, and use the selection tool to draw a rectangle over a grid cell.

The pixel width and height of this cell can be used directly to crop the photo with the Rectangle Select tool. Use the cell width and height to set a fixed aspect ratio of the selection area (the aspect matters, not the specific pixel dimensions).

The spacing has to be adjusted for precise positioning, see below.

```
\SetEvents{September}{%
% All this will be in tikzpicture environment, after drawing the calendar.
%
% This adds a semicolon to end the \calendar command so we can start adding nodes.
\semicolonEndTikzCalendar%
%
% Overlaying a photo with text on a specific date in this month
```

```
%
% photo node
\node (day-photo) [
  above right=3.2pt and 0.1pt of cal09-2018-09-18.north west,
  inner sep=0pt,
  anchor=north west,
] {%
  \includegraphics[width={\@t@calendar@dayXshift - 0.25pt}]{TomBombadil-crop}%
};%
%
% the photo is dark, add a white background for the day number
\node (top-whitecircle) [
  below right=0.3pt and 0.1pt of cal09-2018-09-18-daynumber.center,
  anchor=center,
  circle,
  minimum width=15pt,
  inner sep=0pt,
  fill=white,
  opacity=0.6,
] {};%
%
% the photo is dark, add a white background for the day's event text
\node (bottom-whitebox) [
  above left=0pt and 0pt of day-photo.south west,
  anchor=south west,
  minimum width={\@t@calendar@dayXshift - 0.25pt},
  minimum height=18pt,
  inner sep=0pt,
  fill=white,
  opacity=0.6,
] {};%
%
% repeat the day number (the photo has covered it)
\calDayNumberNode{cal09-2018-09-18}{-redo}{\dayTextFmt 18}%
%
% print the day's marks (moon phases in this case) next to the day number
\parseMonthMarksDayTextInlineUsing{./data/moonphases-2018.csv}%
%
% print the day's event text
\parseMonthInlineNotes%
}{}
```

4.3 Disabling the notepad area

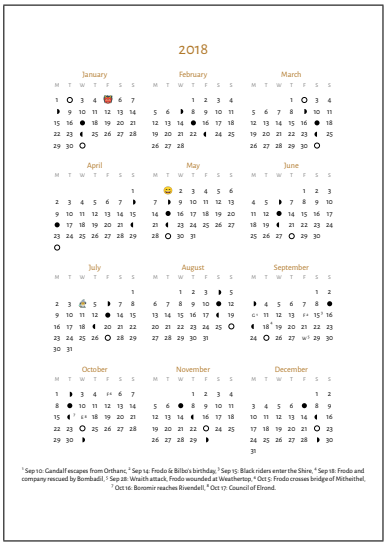
Formatting commands and sizes have to renewed in `\photoAndNotesFmt` (add it to the preamble), because every month page first runs a command which resets them to default values, and then runs the above formatting command to change the defaults.

Disable the notepad area by renewing `\notepadContent` to empty.

In this case the calendar can be taller. Find a month with six rows to experiment by changing `\@t@calendar@minimumHeight` and `\@t@calendar@verticalSpacing`.

```
\renewcommand\photoAndNotesFmt{%
  \renewcommand\notepadContent{}%
  \setlength{\@t@calendar@minimumHeight}{\calPaperHeight - 20mm}%
  \setlength{\@t@calendar@verticalSpacing}{30mm}%
}
```

5 EXAMPLE: YEAR PLANNER PAGE



5.1 DOCUMENT SETUP

cal-year-planner.tex

```
\documentclass[
  year = 2018,
  eventsCsv = ./data/holidays.csv,
  markDefaultsCsv = ./data/mark_defaults.csv,
  imageFolder = ./photos/,
]{wallcalendar}

\makeatletter

\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}

% Use two CSV files for day text input to include the moon phases
\renewcommand\@wall@plm[1]{%
\luadirect{
require("wallcalendar-helpers.lua")
monthMarksDayText(\luastring{#1}, nil, \luastring{\plannerMarksDayTextCSV})
monthMarksDayText(\luastring{#1}, nil, \luastring{./data/moonphases.csv})
tex.sprint(';')
}}
```

5.2 \YEARPLANNERPAGE

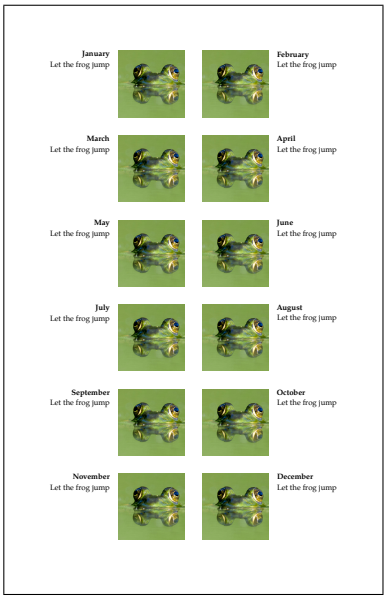
```
\newcommand\plannerYearFmt{\fontsize{26}{26}\selectfont\color{orangegold}}

\newlength\plannerNotesSep
\setlength{\plannerNotesSep}{3mm}

\newcommand\preYearPlannerPageHook{%
  \setlength{\markNumberAbove}{-9pt}%
  \setlength{\markNumberRight}{-6pt}%
  \setlength{\markDayTextAbove}{-11pt}%
  \setlength{\markDayTextRight}{-6pt}%
}
```

```
\newcommand\postYearPlannerPageHook{%
  \setlength{\markNumberAbove}{-10pt}%
  \setlength{\markNumberRight}{-3pt}%
  \setlength{\markDayTextAbove}{-10pt}%
  \setlength{\markDayTextRight}{-3pt}%
}

\newcommand\printPlannerTitle{\plannerYearFmt \CalendarYear}

\newcommand\YearPlannerPage{%
\newpage
\ifvarnishmask
\mbox{}
\else
\preYearPlannerPageHook
{\centering

{\printPlannerTitle}

\vspace*{7mm}

\YearPlannerPortrait

\vspace*{\plannerNotesSep}

\plannerEvents

}
\postYearPlannerPageHook

\fi
}

\makeatother
```

5.3 Use it

```
\begin{document}

\YearPlannerPage

\end{document}
```

6 EXAMPLE: PHOTO THUMBNAILS PAGE



6.1 Document setup

`cal-thumbnails.tex`

```
\documentclass[
  year = 2018,
  imageFolder = ./photos/,
]{wallcalendar}

\makeatletter

\usepackage{fontspec}
\defaultfontfeatures{Ligatures={TeX}}
\setmainfont{TeX Gyre Pagella}
\newfontfamily\dejaVuSans{DejaVu Sans}

\newlength\@wall@tmp@a
\newlength\@wall@tmp@b
```

6.2 \ThumbWithCaptionLeftSide

Typesets the photo thumb image with its caption text on the left side.

```
\ThumbWithCaptionLeftSide{January}
```

```
\newlength\@wall@thumbWidth
\newlength\@wall@thumbHeight
\newlength\@wall@thumbCaptionWidth
\setlength{\@wall@thumbWidth}{0.1749\calPaperWidth}% 30mm at the 6.75in page width, 0.1749 = 1/5.715
\setlength{\@wall@thumbHeight}{\@wall@thumbWidth}
\setlength{\@wall@thumbCaptionWidth}{0.2333\calPaperWidth}% 40mm at 6.75in page width

\newcommand\thumbFmt{}
\newcommand\thumbMonthFmt{\fontsize{10}{13}\selectfont\bfseries}
\newcommand\thumbCaptionFmt{\fontsize{10}{13}\selectfont}

\def\@wall@thumbFile{}
```

```
\def\@wall@photoCaption{}

\newcommand\ThumbWithCaptionLeftSide[1]{%
\pgfkeys{/Photo/#1/thumbFile/.get=\@wall@thumbFile}%
\ifx\@wall@thumbFile\empty
  \pgfkeys{/Photo/#1/file/.get=\@wall@thumbFile}%
\fi
\pgfkeys{/Photo/#1/caption/.get=\@wall@photoCaption}%
% Thumbnail caption
\ifvarnishmask%
\hspace*{\@wall@thumbWidth}
\else%
\begin{minipage}[b][\@wall@thumbHeight][t]{\@wall@thumbCaptionWidth}%
\raggedleft
\thumbFmt
{\thumbMonthFmt \@tr@monthNumName{\monthToNum{#1}}}\par
{\thumbCaptionFmt \@wall@photoCaption}%
\end{minipage}%
\fi%
\hspace*{3mm}
% Thumbnail photo
\begin{minipage}[b][\@wall@thumbHeight]{\@wall@thumbWidth}%
% FIXME placeholder
%\placeholder{%
\includegraphics[ keepaspectratio, height=\@wall@thumbHeight ]{\@wall@thumbFile}%
%}%
\end{minipage}%
}
```

6.3 \ThumbWithCaptionRightSide

Typesets the photo thumb image with its caption text on the right side.

```
\ThumbWithCaptionRightSide{January}
```

```
\newcommand\ThumbWithCaptionRightSide[1]{%
\pgfkeys{/Photo/#1/thumbFile/.get=\@wall@thumbFile}%
\ifx\@wall@thumbFile\empty
  \pgfkeys{/Photo/#1/file/.get=\@wall@thumbFile}%
\fi
\pgfkeys{/Photo/#1/caption/.get=\@wall@photoCaption}%
% Thumbnail photo
\begin{minipage}[b][\@wall@thumbHeight]{\@wall@thumbWidth}%
% FIXME placeholder
%\placeholder{%
\includegraphics[ keepaspectratio, height=\@wall@thumbHeight ]{\@wall@thumbFile}%
%}%
\end{minipage}%
\hspace*{3mm}
% Thumbnail caption
\ifvarnishmask%
\hspace*{\@wall@thumbWidth}
\else%
\begin{minipage}[b][\@wall@thumbHeight][t]{\@wall@thumbCaptionWidth}%
\raggedright
\thumbFmt
{\thumbMonthFmt \@tr@monthNumName{\monthToNum{#1}}}\par
{\thumbCaptionFmt \@wall@photoCaption}%
\end{minipage}%
\fi%
}
```

6.4 \ThumbsPage

```
\newlength{\thumbColumnWidth}
\newlength{\thumbColumnHeight}
\newlength{\thumbSep}
\newlength{\@t@a}
\newlength{\@t@b}

% vertical spacing
\setlength{\@t@a}{0.2\textheight}

% horizontal spacing
\setlength{\@t@b}{0.2\@wall@thumbCaptionWidth}% 8mm at 40mm caption width

\ifdimcomp{\@t@a}{<}{\@t@b}{
  \setlength{\thumbSep}{\@t@a}
}{
  \setlength{\thumbSep}{\@t@b}
}

\setlength{\thumbColumnWidth}{0.5\calPaperWidth - \thumbSep}
\setlength{\thumbColumnHeight}{\textheight}

\newcommand\ThumbsPage{%
\clearpage

\hspace*{-\@wall@leftMargin}%
\hspace*{-1pt}% small correction, space gets in somewhere
% Wrap
\begin{minipage}[t][\thumbColumnHeight]{\calPaperWidth}%
\centering%
\setlength{\parindent}{0pt}%
\setlength{\parskip}{0pt}%

% NOTE: multicols will not \vfill
% Left Column
\begin{minipage}[t][\thumbColumnHeight]{\thumbColumnWidth}%
\raggedleft

\ThumbWithCaptionLeftSide{January}

\vspace*{\thumbSep}

\ThumbWithCaptionLeftSide{March}

\vspace*{\thumbSep}

\ThumbWithCaptionLeftSide{May}

\vspace*{\thumbSep}

\ThumbWithCaptionLeftSide{July}

\vspace*{\thumbSep}

\ThumbWithCaptionLeftSide{September}

\vspace*{\thumbSep}

\ThumbWithCaptionLeftSide{November}

% End of left column
\end{minipage}%
```

```
%
\hspace*{\thumbSep}%
%
% Right column
\begin{minipage}[t][\thumbColumnHeight]{\thumbColumnWidth}%
\raggedright

\ThumbWithCaptionRightSide{February}

\vspace*{\thumbSep}

\ThumbWithCaptionRightSide{April}

\vspace*{\thumbSep}

\ThumbWithCaptionRightSide{June}

\vspace*{\thumbSep}

\ThumbWithCaptionRightSide{August}

\vspace*{\thumbSep}

\ThumbWithCaptionRightSide{October}

\vspace*{\thumbSep}

\ThumbWithCaptionRightSide{December}

% End of right column
\end{minipage}%

% End of wrap
\end{minipage}%
}
```

6.5 Setup the photo keys

```
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{January}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{February}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{March}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{April}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{May}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{June}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{July}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{August}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{September}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{October}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{November}
\SetPhoto[file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{December}

\makeatother
```

6.6 Use it

```
\begin{document}

\ThumbsPage

\end{document}
```

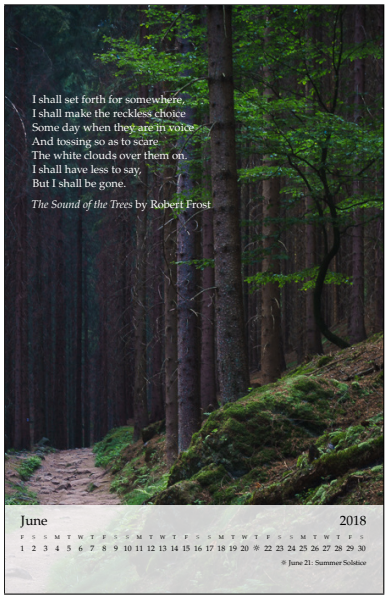
7 Examples from doc/examples folder

7.1 Portrait page layouts

`doc/examples/cal-plain.tex` demonstrates three different `\MonthPage` layouts using portrait-oriented pages. Each month in the example uses a different layout and photo placement option.

7.1.1 Full page photo, calendar days overlaid with opacity

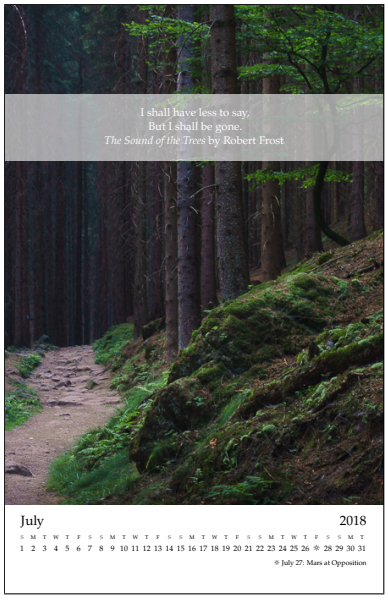
```
\MonthPage[layout=full page, put photo=full page]{June}
```



The photo covers the entire page (with bleed). Calendar days are overlaid at the bottom with a semi-transparent background.

7.1.2 Full page photo, photo above the calendar days

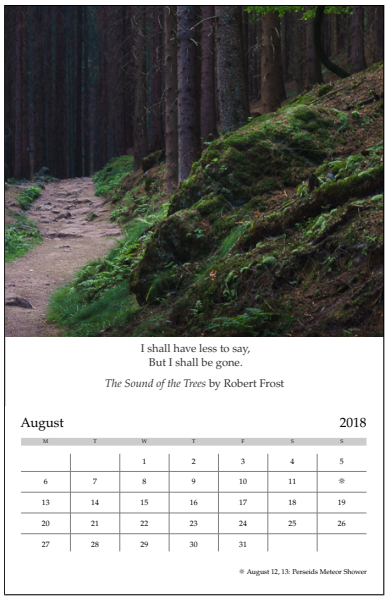
```
\MonthPage[layout=full page, put photo=full width above calendar]{July}
```



The photo fills the width of the page and sits above the calendar row. The calendar days are drawn on a solid (opaque) background below the photo.

7.1.3 Small landscape photo, with a calendar grid

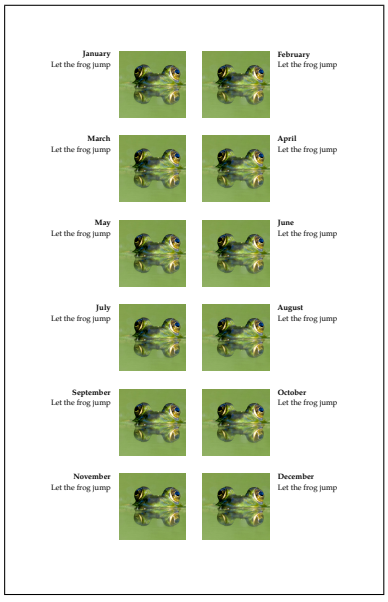
```
\MonthPage[layout=small landscape, put photo=full width]{August}
```



A landscape-oriented photo is placed at the top, scaled to the full page width. Below it, a calendar grid shows the month with day-of-week headers and room for events.

7.1.4 Thumbnails and captions page

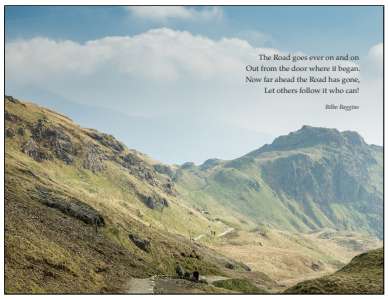
```
doc/examples/cal-thumbnails.tex
```

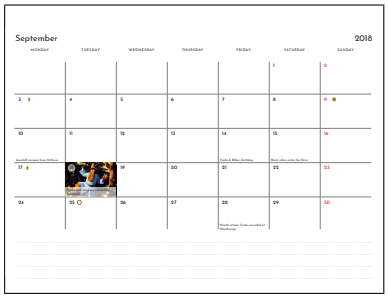


A standalone page showing thumbnail images for all twelve months arranged in two columns, with captions alongside each thumbnail. See the Example: Photo Thumbnails Page section for details.

7.2 Landscape photo and notes on two pages

`doc/examples/cal-photo-and-notes.tex`

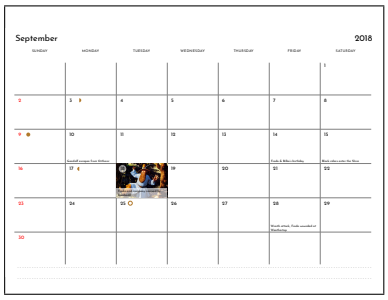




This layout uses the `layout=photo and notes` option. Each month produces two pages: a landscape photo page and a facing calendar page with a grid, event marks, and a notepad area. The example also includes a title page using `\TitlePage`.

7.2.1 *Sunday-first week layout*

`doc/examples/cal-photo-and-notes-sunday.tex`



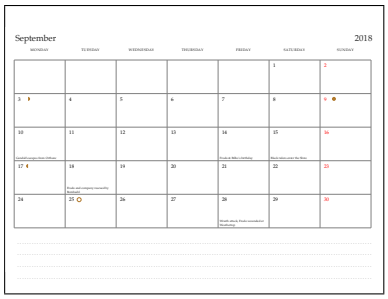
The `sundayFirst` class option shifts the week to start on Sunday instead of Monday. All layouts (monthly grids, year planners, compact planners) respect this option. The day-of-week headings automatically reorder to show Sunday in the first column.

This example is based on the Photo and Notes layout, showing September with Sunday as the first day of the week.

```
\documentclass[
  year = 2018,
  language = english,
  imageFolder = ./photos/,
  sundayFirst,
]{wallcalendar}
```

7.2.2 No photos, with notes area

`doc/examples/cal-no-photos.tex`



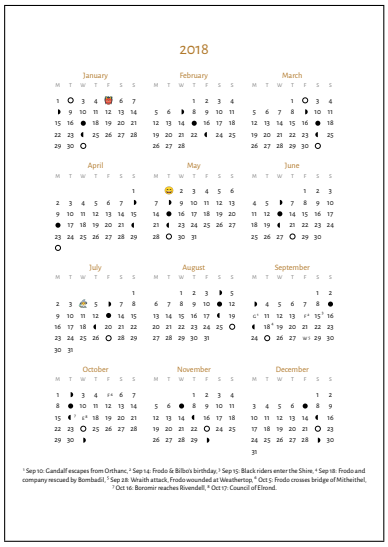
This uses the same `photo and notes` layout but with `disabled = true` on all photos. This produces calendar-only pages with a grid, event marks and notepad area – suitable for a desk planner or scheduling tool.

7.3 Year planners

The planners support event marks from CSV files and moon phase symbols via the `eventsCsv`, `moonsCsv` and `markDefaultsCsv` class options.

7.3.1 Portrait year planner

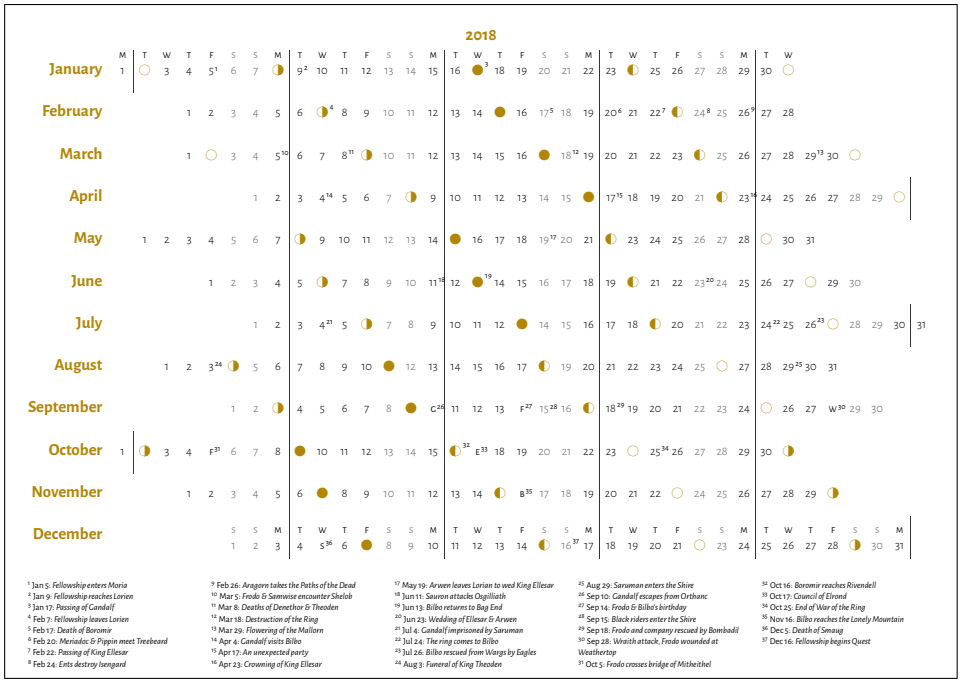
`doc/examples/cal-year-planner.tex`



Uses `\YearPlannerPortrait` to render a 3-column, 4-row grid of monthly calendars in portrait orientation. This example defines a custom `\YearPlannerPage` command that wraps `\YearPlannerPortrait` with a title and `\plannerEvents` footer. The `\plannerEvents` command prints all events from the CSV that have a footnote marker.

7.3.2 Landscape year planner with dates in rows

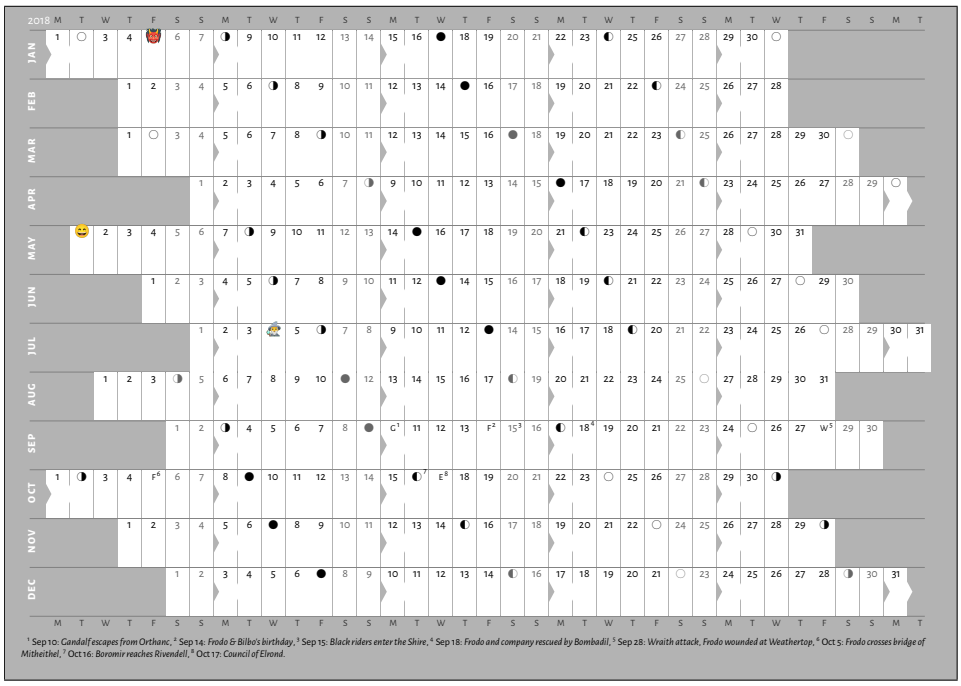
`doc/examples/cal-year-planner-rows.tex`



Uses `\YearPlannerLandscape` to render the entire year as a continuous row-based layout in landscape orientation. Event footnotes are printed below in a `multicols` environment using `\parseYearEvents`.

7.3.3 *Landscape year planner with grid*

`doc/examples/cal-year-planner-grid.tex`

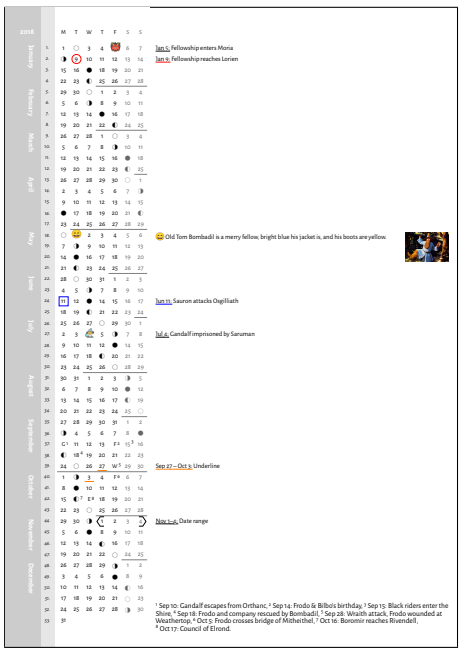


Uses `\YearPlannerLandscapeGridPage` which renders a grid-style planner in landscape orientation. The layout includes event footnotes at the bottom.

Uses the `a4PaperLandscape` page preset with the `dateAnnotations = false` option to disable annotation superscripts on dates (event footnotes are shown separately).

7.3.4 *Compact year planner with weekly notes*

`doc/examples/cal-year-planner-compact.tex`



Uses `\YearPlannerCompact` to render a compact year planner in portrait (A4) orientation with the calendar dates on the left side of the page and annotated weekly notes alongside. This layout also supports weekly images positioned next to the notes using `\plannerWeeklyNotes` and `\plannerWeeklyImages`.

Uses the `a4Paper` page preset. The `\plannerMoonFormat` and `\monthMarkFmt` commands can be renewed to customize moon phase symbol size and event mark formatting.

7.4 Generating moon phase data

The year planner examples include pre-generated moon phase CSV files (e.g. `data/moonphases-2018.csv`). To generate moon phases for an arbitrary year, use the `generate_moon_phases.py` script in `doc/examples/scripts/`.

The script uses PyEphem for astronomically accurate moon phase calculations and outputs a CSV file in the `moonsCsv` format expected by wallcalendar (with `date`, `day_text`, and `footnote` columns).

7.4.1 Setup

The script uses Poetry for dependency management. From the `scripts` directory:

```
cd doc/examples/scripts
poetry install
```

The `poetry.toml` file has `virtualenvs.in-project = true` so the virtual environment is created as `scripts/.venv/`.

7.4.2 Usage

```
poetry run python generate_moon_phases.py YEAR [OUTPUT_FILE]
```

To generate moon phases for use with the year planner examples:

```
cd doc/examples/scripts
poetry run python generate_moon_phases.py 2026 ../data/moonphases-2026.csv
```

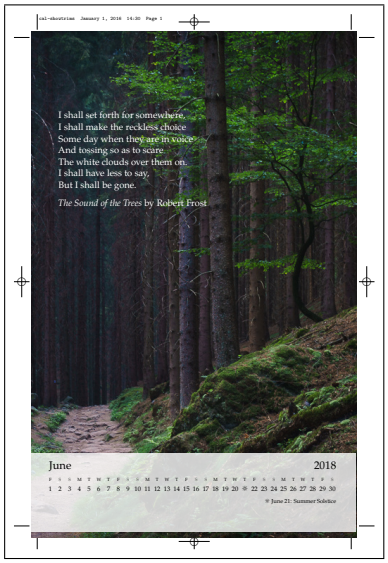
Then reference the generated file in your `.tex` document:

```
\documentclass[
  year = 2026,
  eventsCsv = ./data/holidays-compact.csv,
  moonsCsv = ./data/moonphases-2026.csv,
  markDefaultsCsv = ./data/mark_defaults.csv,
]{wallcalendar}
```

If no output file is specified, the script writes to `moonphases-YEAR.csv` in the current directory.

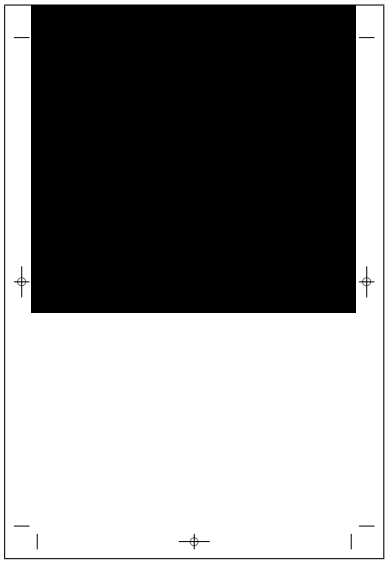
7.5 Class options for printing and debugging

7.5.1 Showtrims

`doc/examples/cal-showtrims.tex`

The `showtrims` class option adds crop-marks around the page, showing where the paper should be trimmed after printing. This is essential when using bleeded images that extend beyond the trim area.

7.5.2 Varnishmask

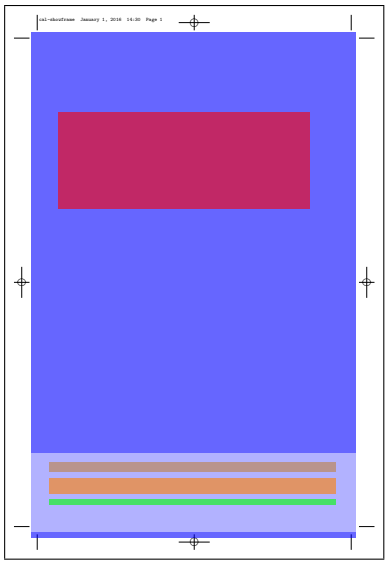
`doc/examples/cal-varnishmask.tex`



The `varnishmask` class option generates a UV mask (varnish mask) file. This produces black shapes where the photos are, on a white background. The mask file is used by the print shop to apply selective gloss varnish on the photo areas.

7.5.3 Showframe

`doc/examples/cal-showframe.tex`



The `showframe` class option draws the page structure: margins, text area, and content box boundaries. Useful for debugging layout positioning when designing custom formatting or adjusting element placement.

8 DOCUMENTCLASS OPTIONS

Options are set as key-value pairs in the `\documentclass` call:

```
\documentclass[
  year = 2018,
  language = english,
  imageFolder = ./photos/,
  eventsCsv = ./data/holidays.csv,
  moonsCsv = ./data/moonphases-2018.csv,
  markDefaultsCsv = ./data/mark_defaults.csv,
  a4Paper,
  hMargin = 7mm,
  topMargin = 7mm,
  bottomMargin = 7mm,
]{wallcalendar}
```

year = 2018 The calendar year (e.g. 2018)

language = english Language for translations (e.g. english, japanese, hungarian)

imageFolder = ./photos/ Path to the folder containing photo images

eventsCsv = ./data/holidays.csv Path to the CSV file with calendar events

moonsCsv = ./data/moonphases-2018.csv Path to the CSV file with moon phase data

markDefaultsCsv = ./data/mark_defaults.csv Path to the CSV file with default mark styles

translationsInputFile = frog-english.tex Path to a file with `\SetTxt{}` translation keys

paperWidth = 11in - 15mm Custom paper width (default: US Letter landscape width)

paperHeight = 8.5in - 15mm Custom paper height (default: US Letter landscape height)

hMargin = 7mm Horizontal margin on both sides (default: 17mm)

topMargin = 15mm Top margin (default: 20.5mm)

bottomMargin = 7mm Bottom margin (default: 10mm)

trimSpace = 15mm Extra stock size for trim marks (default: 35mm)

a4Paper Preset for A4 portrait (210mm x 297mm)

a4PaperLandscape Preset for A4 landscape (297mm x 210mm)

showframe Show page structure for debugging layout (boolean, default: false)

showtrims Show crop marks for printing (memoir class option, omit to disable)

varnishmask Generate UV varnish mask pages (boolean, default: false)

gridSurround Draw grid lines around calendar day cells (boolean, default: false)

sundayFirst Start the week on Sunday instead of Monday (boolean, default: false)

dateAnnotations = true Enable/disable date annotation superscripts (string, default: true)

Note: `dateAnnotations` is stored as a string (not a TEX boolean) because its value is passed to Lua code via `\luadirect` where it is compared as `isEqual(..., "true")`. This means it only recognizes the literal string `"true"` to enable annotations; any other value (including `"false"`) disables them.

9 USER COMMANDS

```
\SetPhoto[ options ]{ month }
```

Set the photo for a given month. Options include `file` (image filename without extension), `bleed` (bleed amount), `thumbFile` (thumbnail image for the thumbnails page), `caption` (caption text), `disabled` (skip the photo), and `yOffset` (vertical offset).

```
\SetPhoto[bleed=3mm, file={obscure-crop}]{June}
\SetPhoto[bleed=3mm, file={frog-crop}, thumbFile={frog-crop-thumb}, caption={Let the frog jump}]{January}
\SetPhoto[disabled = true]{September}
```

```
\SetQuote[ options ]{ month }{ quote text }
```

Set a quote overlay for a given month. Options include `xOffset` and `yOffset` to position the quote minipage relative to the top-left corner of the page.

```
\SetQuote[xOffset=-5mm, yOffset=-20mm]{June}{%
  \raggedright\Large\color{white}%
  I shall be gone.%
}
```

```
\SetCalendar[ options ]{ month }
```

Set calendar rendering options for a given month. The main option is `bg/.style` which controls the TikZ style of the calendar background.

```
\SetCalendar[bg/.style={opacity=0.8}]{June}
\SetCalendar[]{September}
```

```
\SetEvents[ options ]{ month }{ calendar tikz marks }{ events text }
```

Set event marks and event text for a given month. The second argument contains TikZ calendar mark expressions. The third argument contains the text to display in the events area.

```
\SetEvents{June}{%
  if (equals=2018-06-21)
    [day text={\dejaVuSans\char"263C}];% U+263C white sun with rays
}{%
  \raggedleft
  {\dejaVuSans\char"263C} June 21: Summer Solstice
}
```

```
\MonthPage[ options ]{ month }
```

Typeset a month page. The `layout` option selects the layout handler (`full page`, `small landscape`, or `photo and notes`). The `put photo` option controls photo placement (`full page`, `full width above calendar`, or `full width`).

```
\MonthPage[layout=full page, put photo=full page]{June}
\MonthPage[layout=full page, put photo=full width above calendar]{July}
\MonthPage[layout=small landscape, put photo=full width]{August}
\MonthPage[layout=photo and notes, put photo=full page]{September}
```

10 PAGE LAYOUT

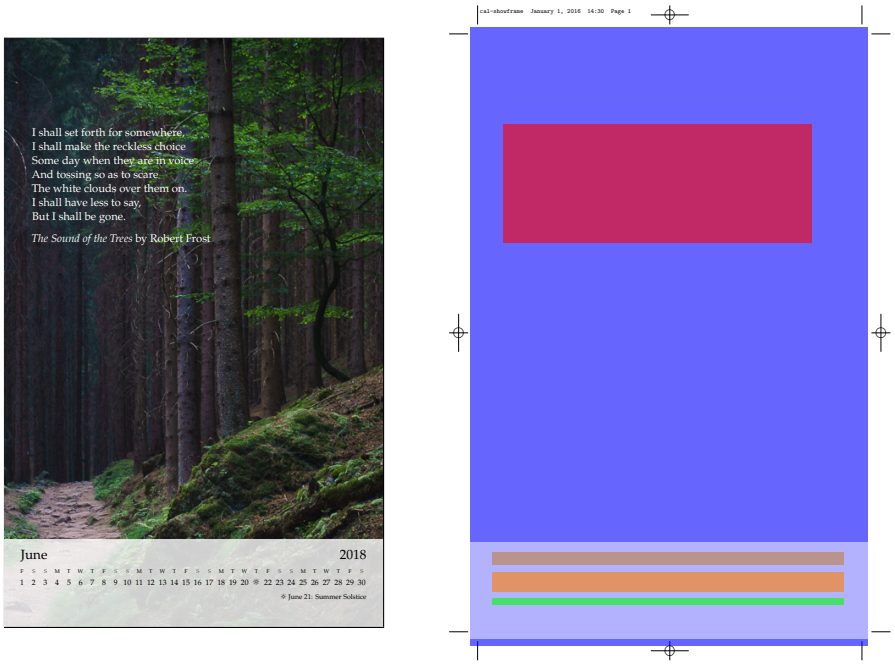
We will review how the page layout happens when we call `\MonthPage`.

Before calling `\MonthPage` to typeset a particular month, be sure that the content for the given month has been already set with `\SetPhoto`, `\SetQuote`, `\SetCalendar` and `\SetEvents`.

Then we can typeset that month, for example:

```
\MonthPage[layout=full page, put photo=full page]{June}
```

This will be a full page portrait month with the photo scaled to cover the entire page with additional bleed on all four sides.



The above `showframe` figure shows the structural elements of the page.

Every layout is implemented by a single handler macro which will deal with all the typesetting of the given page. The `full page` key is set to the `\@wall@fullPageLayout` macro by default, and so this gets called.

The layout macro is just a free-style placeholder. It can access the photo, quote, calendar and events as set earlier, but it is up to the macro to implement what to do with them.

This is for the convenience of setting the page elements using the same interface, but being able to execute different layouts for different pages.

The class contains two layout examples. The `full page` layout is best for portrait photos that can be scaled to cover the entire page. The `small landscape` layout is for landscape photos which can be scaled horizontally, possibly bleeding into the side margins.

10.1 Formatting Hooks: \fullPageFmt

Each layout runs a formatting command before typesetting the page. The `full page` layout calls `\fullPageFmt`, which sets hooks for fonts, sizes, colors and alignment. You can `\renewcommand` it in your preamble to customize the appearance.

The default definition is:

```
\newcommand\fullPageFmt{%
  \renewcommand*\monthFmt{\LARGE}%
  \renewcommand*\yearFmt{\LARGE}%
  \renewcommand*\dayLetterColor{}%
  \renewcommand*\dayLetterFmt{\tiny}%
  \renewcommand*\dayTextFmt{\small}%
  \renewcommand*\quoteFmt{}%
  \renewcommand*\headingFmt{\centering}%
  \renewcommand*\calendarFmt{\centering}%
  \renewcommand*\eventsFmt{%
    \setlength{\parindent}{0pt}\raggedleft\footnotesize%
  }%
}
```

The hooks control the following elements:

- `\monthFmt` – Font and size for the month name
- `\yearFmt` – Font and size for the year number
- `\dayLetterColor` – Color applied to the day-of-week letter row
- `\dayLetterFmt` – Font and size for day-of-week letters (M, T, W…)
- `\dayTextFmt` – Font and size for the day numbers in the calendar
- `\quoteFmt` – Formatting applied to the quote area
- `\headingFmt` – Alignment for the month/year heading (e.g. `\centering`)
- `\calendarFmt` – Alignment for the calendar area
- `\eventsFmt` – Formatting for the events text below the calendar

See the Translations tutorial for an example of renewing `\fullPageFmt` to use a Japanese font.

10.2 Formatting Hooks: \smallLandscapeFmt

The `small landscape` layout calls `\smallLandscapeFmt`. In addition to the same hooks as `\fullPageFmt`, it also sets lengths that control the calendar grid dimensions.

The default definition is:

```
\newcommand\smallLandscapeFmt{%
  \renewcommand*\monthFmt{\LARGE}%
  \renewcommand*\yearFmt{\LARGE}%
  \renewcommand*\dayLetterColor{}%
  \renewcommand*\dayLetterFmt{\tiny}%
  \renewcommand*\dayTextFmt{\small}%
  \renewcommand*\quoteFmt{\centering}%
  \renewcommand*\headingFmt{\centering}%
  \renewcommand*\calendarFmt{\centering}%
  \renewcommand*\eventsFmt{%
    \setlength{\parindent}{0pt}\raggedright\footnotesize%
  }%
  \setlength{\@t@calendar@minimumHeight}{85mm}%
  \setlength{\@t@calendar@hmargin}{20pt}%
  \setlength{\@t@calendar@verticalSpacing}{20mm + 5mm + 6.5mm}%
}
```

The additional lengths are:

- `\@t@calendar@minimumHeight` – Minimum height of the calendar grid area
- `\@t@calendar@hmargin` – Horizontal margin inside the calendar grid
- `\@t@calendar@verticalSpacing` – Vertical space reserved for the day headings, events area, and white space below the grid

10.3 Formatting Hooks: \photoAndNotesFmt

The `photo and notes` layout calls `\photoAndNotesFmt`. See the Example: Photo and Notes with Title page section for details on customizing this layout, including how to disable the notepad area.

11 CONTACT

Github: https://github.com/profound-labs/wallcalendar/

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